

1. We recall that the limit $\lim_{(x,y) \rightarrow (0,0)} f(x,y)$ using the polar coordinates r, θ with $x = r \cos \theta$ and $y = r \sin \theta$, is obtained as follows.

- (i) We first find the limit along some known lines passing through the point $(0,0)$ in order to identify the candidate limit L .
- (ii) We estimate $|f(r \cos \theta, r \sin \theta) - L| \leq g(r)$, where g is a function of r only with the limit $\lim_{r \rightarrow 0^+} g(r) = 0$. The function g is obtained through our knowledge of the properties of the trigonometric functions sine and cosine.
- (iii) We then say that $\lim_{r \rightarrow 0^+} f(r \cos \theta, r \sin \theta) = L$ uniformly in θ which is equivalent to saying that $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = L$.

Find the limit $\lim_{(x,y) \rightarrow (0,0)} f(x,y)$ for:

(a) $f(x,y) := \frac{\ln(1+xy)}{\sqrt{x^2+y^2}};$

We find that the limit along both axes is equal to 0. Candidate $L = 0$?
Along the line $y = x$, we find that

$$f(x,x) = \frac{\ln(1+x^2)}{|x|\sqrt{2}}$$

Now, we find that

$$\begin{cases} \lim_{x \rightarrow 0^+} f(x,x) = \lim_{x \rightarrow 0^+} \frac{\ln(1+x^2)}{x\sqrt{2}} = \lim_{x \rightarrow 0^+} \frac{2x}{(1+x^2)\sqrt{2}} = 0, \\ \lim_{x \rightarrow 0^-} f(x,x) = \lim_{x \rightarrow 0^-} \frac{\ln(1+x^2)}{-x\sqrt{2}} = \lim_{x \rightarrow 0^-} \frac{-2x}{(1+x^2)\sqrt{2}} = 0 \end{cases} \Rightarrow \lim_{x \rightarrow 0} f(x,x) = 0.$$

We could also write

$$f(x,x) = \frac{\ln(1+x^2)}{|x|\sqrt{2}} = \frac{\ln(1+x^2)}{x^2} \times \frac{|x|}{\sqrt{2}}$$

We use the fact that $\lim_{t \rightarrow 0} \frac{\ln(1+t)}{t} = 1$ to get $\lim_{x \rightarrow 0} \frac{\ln(1+x^2)}{x^2} = 1$ and hence, $\lim_{x \rightarrow 0} f(x,x) = 0$.

Now, to check whether the limit is equal to 0 we write

$$f(x,y) = \frac{\ln(1+xy)}{xy} \times \frac{xy}{\sqrt{x^2+y^2}}$$

As mentioned above, we find $\lim_{(x,y) \rightarrow (0,0)} \frac{\ln(1+xy)}{xy} = 1$. To find the limit $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{\sqrt{x^2+y^2}}$ we use polar coordinates as follows

$$0 \leq \frac{r^2 |\cos \theta \sin \theta|}{\sqrt{r^2 \cos^2 \theta + r^2 \sin^2 \theta}} = r |\cos \theta \sin \theta| \leq r \rightarrow 0.$$

Hence, $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{\sqrt{x^2+y^2}} = 0$ and therefore,

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = \lim_{(x,y) \rightarrow (0,0)} \frac{\ln(1+xy)}{xy} \times \lim_{(x,y) \rightarrow (0,0)} \frac{xy}{\sqrt{x^2+y^2}} = 1 \times 0 = 0.$$

(b) $f(x,y) := \frac{x^2 y^3}{x^4 + y^4}.$

We find that the limit along both axes is equal to 0. Also, along the line $y = x$, we find that $f(x,x) = x/2 \rightarrow 0$ as $x \rightarrow 0$. Candidate $L = 0$?

Now we consider

$$|f(r \cos \theta, r \sin \theta) - 0| = \frac{r^5 |\cos^2(\theta) \sin^3(\theta)|}{r^4 (\cos^4(\theta) + \sin^4(\theta))} = \frac{r |\cos^2(\theta) \sin^3(\theta)|}{\cos^4(\theta) + \sin^4(\theta)} \leq \frac{r}{\frac{1}{2}} = 2r \rightarrow 0 \text{ as } r \rightarrow 0^+$$

where we have used $|\cos^2(\theta) \sin^3(\theta)| \leq 1$ and $\cos^4(\theta) + \sin^4(\theta) \geq 1/2$ from the hint.

Hence, $\lim_{r \rightarrow 0^+} f(r \cos \theta, r \sin \theta) = 0$ uniformly in θ which is equivalent to saying that

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0.$$

2. Consider the function $f(x,y) := \begin{cases} \frac{\arcsin(x^2+y^2)}{\sqrt{x^2+y^2}} & \text{if } (x,y) \neq (0,0), \\ 0 & \text{if } (x,y) = (0,0). \end{cases}$

(a) Find the domain of the function f .

Knowing that the function $\arcsin t$ is defined for $-1 \leq t \leq 1$, we get that $f(x,y)$ is defined for every (x,y) such that

$$-1 \leq x^2 + y^2 \leq 1 \quad \Leftrightarrow \quad x^2 + y^2 \leq 1.$$

Hence, the domain $\text{Dom}(f)$ of the function f is the unit disc centered at the origin $(0,0)$.

(b) Study the continuity of f at the point $(0,0)$.

We want to check whether $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0$.

Along the x -axis, y -axis and the line $y = x$, we get

$$f(x,0) = \frac{\arcsin(x^2)}{|x|}, \quad f(0,y) = \frac{\arcsin(y^2)}{|y|}, \quad f(x,x) = \frac{\arcsin(2x^2)}{|x|\sqrt{2}}$$

whose limits are obtained through the knowledge of the limit $\lim_{t \rightarrow 0} \frac{\arcsin(2t^2)}{t}$.

We recall that $\lim_{t \rightarrow 0} \frac{\arcsin(2t^2)}{t} = 0$ using L'H. Thus, we find that the limit along both axes and the line $y = x$ is equal to 0. So, the candidate limit is $L = 0$. Using polar coordinates, we get

$$|f(r \cos \theta, r \sin \theta) - 0| = \frac{\arcsin(r^2)}{r} \longrightarrow 0 \text{ as } r \rightarrow 0^+.$$

Hence, $\lim_{r \rightarrow 0^+} f(r \cos \theta, r \sin \theta) = 0$ uniformly in θ which is equivalent to saying that $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0$.

(c) Find, if they exist, $f_x(0, 0)$ and $f_y(0, 0)$.

$$\frac{f(x, 0) - f(0, 0)}{x} = \frac{\arcsin(x^2)}{x|x|} = \begin{cases} \frac{\arcsin(x^2)}{x^2} \rightarrow 1 & \text{as } x \rightarrow 0^+ \text{ by L'H,} \\ \frac{\arcsin(x^2)}{-x^2} \rightarrow -1 & \text{as } x \rightarrow 0^- \text{ by L'H.} \end{cases}$$

Thus $f_x(0, 0)$ does not exist and by symmetry also $f_y(0, 0)$ does not exist.

(d) Study the differentiability of the function f at $(0, 0)$.

It follows from (c) that the function f is not differentiable at the point $(0, 0)$.

3. Let $f(x, y) = x^2y^3$.

- (i) Show that vector $\vec{n} = (2, 3, -1)$ is perpendicular to the tangent plane to the graph of f at the point $(1, 1, 1)$.

The vector $\vec{n}$ is perpendicular to the tangent plane to the graph of f at the point $(1, 1, 1)$ if and only $\vec{n}$ is parallel to the normal vector to the tangent plane. From the lecture notes, we know that the normal vector of the tangent plane to the graph of f at the point $(1, 1, 1)$ is given by

$$\boxed{(-f_x(1, 1), -f_y(1, 1), 1)}.$$

Notice that

$$\begin{aligned} \begin{cases} f_x(x, y) = 2xy^3, \\ f_y(x, y) = 3x^2y^2 \end{cases} &\Rightarrow \begin{cases} f_x(1, 1) = 2, \\ f_y(1, 1) = 3 \end{cases} \\ &\Rightarrow (-f_x(1, 1), -f_y(1, 1), 1) = (-2, -3, 1) = -(2, 3, -1). \end{aligned}$$

Hence, the two vectors are indeed parallel.

- (ii) Find the points (x, y, z) on the graph of f at which the tangent plane is perpendicular to the vector $\vec{u} = (-1, 3, 1)$

(Hint: What is the expression of the normal vector to the tangent plane at a point (x, y, z) of the graph?).

Here, we seek points (x, y, z) on the graph of f , that is, $(x, y, f(x, y))$ such that the vectors $(-f_x(x, y), -f_y(x, y), 1)$ and $(-1, 3, 1)$ are parallel. That is,

$$\boxed{(-f_x(x, y), -f_y(x, y), 1) \parallel (-1, 3, 1) \Leftrightarrow (-f_x(x, y), -f_y(x, y), 1) \times (-1, 3, 1) = \vec{0}}.$$

Notice that

$$\begin{aligned} (-f_x(x, y), -f_y(x, y), 1) \times (-1, 3, 1) &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2xy^3 & -3x^2y^2 & 1 \\ -1 & 3 & 1 \end{vmatrix} \\ &= (-3x^2y^2 - 3, -2xy^3 - 1, -6xy^3 - 3x^2y^2) \end{aligned}$$

So,

$$(-f_x(x, y), -f_y(x, y), 1) \times (-1, 3, 1) = (0, 0, 0) \Leftrightarrow \begin{cases} -3x^2y^2 - 3 = 0, \\ 2xy^3 - 1 = 0, \\ -6xy^3 - 3x^2y^2 = 0 \end{cases}$$

which is impossible from the first equation. Therefore, there are no points on the graph of f at which the tangent plane is perpendicular to the vector $\vec{u} = (-1, 3, 1)$!

If you were asked to find points on the graph of f at which the tangent plane is rather perpendicular to the vector $\vec{u} = (-1, 3, -1)$, you would have obtained

$$(-f_x(x, y), -f_y(x, y), 1) \times (-1, 3, -1) = (3x^2y^2 - 3, 2xy^3 - 1, -6xy^3 - 3x^2y^2).$$

So,

$$\begin{aligned}
& (-f_x(x, y), -f_y(x, y), 1) \times (-1, 3, -1) = (0, 0, 0) \\
& \Leftrightarrow \begin{cases} 3x^2y^2 - 3 = 0, \\ -2xy^3 - 1 = 0, \\ -6xy^3 - 3x^2y^2 = 0 \end{cases} \Leftrightarrow \begin{cases} x^2y^2 = 1, \\ 2xy^3 = -1, \\ 2xy^3 = -x^2y^2 \end{cases} \Leftrightarrow \begin{cases} x^2y^2 = 1, \\ 2xy^3 = -1, \\ 2y = -x \end{cases} \\
& \Rightarrow \begin{cases} x = -2y, \\ 4y^4 = 1 \end{cases} \Rightarrow (x, y) = (-\sqrt{2}, 1/\sqrt{2}), (\sqrt{2}, -1/\sqrt{2}).
\end{aligned}$$

So, we would have found the points $(-\sqrt{2}, 1/\sqrt{2}, 1/\sqrt{2})$ and $(\sqrt{2}, -1/\sqrt{2}, -1/\sqrt{2})$.

4. Let $\lambda \in \mathbb{R}$. Consider the function $f(x, y) := \begin{cases} \frac{x^2 \sin y}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0), \\ \lambda & \text{if } (x, y) = (0, 0). \end{cases}$

(a) Find the value of the parameter λ such that the function f is continuous at $(0, 0)$.

The function f is continuous at $(0, 0)$ if $\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = \lambda$.

We find that $f(x, 0) = f(0, y) = 0$. On the other hand,

$$f(x, x) = \frac{x^2 \sin x}{2x^2} = \frac{\sin x}{2} \rightarrow 0 \text{ as } x \rightarrow 0.$$

Candidate $L = 0$? Notice that

$$|f(x, y) - 0| = \underbrace{\left[\frac{x^2}{x^2 + y^2} \right]}_{\leq 1} |\sin y| \leq |\sin y| \rightarrow 0.$$

We find by the squeeze theorem that $\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = 0$ and thus we get $\lambda = 0$.

(b) Is the function f differentiable at $(0, 0)$ for the value of λ obtained in (a)?

Having $f(x, 0) = f(0, y) = 0$ we find that $f_x(0, 0) = f_y(0, 0) = 0$. Now, it remains to check whether

$$\lim_{(x, y) \rightarrow (0, 0)} \frac{f(x, y) - f(0, 0) - f_x(0, 0)(x - 0) - f_y(0, 0)(y - 0)}{\sqrt{(x - 0)^2 + (y - 0)^2}} = 0.$$

$$\frac{f(x, y) - f(0, 0) - f_x(0, 0)(x - 0) - f_y(0, 0)(y - 0)}{\sqrt{(x - 0)^2 + (y - 0)^2}} = \frac{x^2 \sin y}{(x^2 + y^2)^{3/2}} \rightarrow 0 \text{ along the axes.}$$

However, along the line $y = x$, we get

$$\lim_{x \rightarrow 0^+} \frac{x^2 \sin x}{2\sqrt{2}|x|^3} = \lim_{x \rightarrow 0^+} \frac{1}{2\sqrt{2}} \frac{\sin x}{x} = \frac{1}{2\sqrt{2}} \neq 0.$$

Thus the function f is not differentiable at the point $(0, 0)$.

5. Given the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $f(x, y) = y \arctan(x + y)$ for every $(x, y) \in \mathbb{R}^2$.

- (a) Write down the cartesian equation of the tangent plane to the graph of f at the point $(0, -1)$.

The tangent plane has cartesian equation

$$z = f(0, -1) + f_x(0, -1)(x - 0) + f_y(0, -1)(y + 1).$$

So, we need

$$\begin{cases} f(0, -1) = -\arctan(-1) = \arctan 1 = \frac{\pi}{4}, \\ f_x(x, y) = \frac{y}{1 + (x + y)^2} \Rightarrow f_x(0, -1) = -\frac{1}{2}, \\ f_y(x, y) = \arctan(x + y) + \frac{y}{1 + (x + y)^2} \Rightarrow f_y(0, -1) = -\frac{\pi}{4} - \frac{1}{2} = -\frac{\pi + 2}{4}, \end{cases}$$

Therefore, we obtain the equation

$$z = \frac{\pi}{4} - \frac{1}{2}x - \frac{\pi + 2}{4}(y + 1)$$

- (b) Write down the linear approximation of the function f around the point $(0, -1)$.

$$f(x, y) \approx f(0, -1) + f_x(0, -1)(x - 0) + f_y(0, -1)(y + 1) \text{ for } (x, y) \text{ close enough to } (0, 0).$$

That is,

$$f(x, y) \approx \frac{\pi}{4} - \frac{1}{2}x - \frac{\pi + 2}{4}(y + 1) \text{ for } (x, y) \text{ close enough to } (0, 0).$$

- (c) Deduce from (b) an approximation of the number $f(-0.09, -0.9)$.

We deduce that

$$f(-0.09, -0.9) \approx \frac{\pi}{4} + \frac{0.09}{2} - \frac{0.1(\pi + 2)}{4}.$$